



DDOL Youth Workforce Connect

One system for every youth workforce opportunity in Delaware

An AI pipeline that reads scattered partner websites and turns them into one clean, trustworthy calendar

Team name: Division of Labor

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THE PROBLEM

130+ partner organizations in 130+ calendars

Who experiences it

Parents, youth, case workers, and the 134 organizations that run Delaware youth programs

What happens today

Every organization posts its own events on its own website, with no shared view across the website

Why it matters

Families have to check dozens of sites.
Now they only need one

130+

Delaware youth-serving organizations,
each maintaining its own events page

130+

Different calendars for each one



Current Organizations

39

Community Partners

33

Schools

19

Workforce
Development

22

Government

9

Churches

8

Employers

4

Summer Programs



THE SOLUTION

An AI pipeline reads every partner site, **extracts, finds duplicates and flags them as active or inactive**-- creating one trustable calendar



Partner site

134 organization
event pages, no
shared format



Scraper

Fetches + cleans
each page's text



AI extraction

The AI reads it,
and outputs the
correct data



Duplicates

Finds duplicates,
and doesn't
consider them



Public calendar

Search + filter,
mobile-first

Why an LLM and not per-site scrapers? Partner pages have zero consistent structure. Some are full calendars while some are one paragraph of news.



WHAT WE BUILT

A six-stage pipeline, plus two working frontends

- **Ingestion**
Loads organization data into DynamoDB. User-submitted organizations enter a separate admin review queue.
- **Scraper**
AWS Lambda automatically crawls approved organization websites and discovers relevant event pages.
- **AI extraction**
Amazon Bedrock extracts structured event details including name, dates, location, age range, cost, description, and event link.
- **De Duplicate Engine**
Generates a deterministic event identity using organization, event name, location, and start date to prevent duplicate events.
- **Admin Review**
Scraped events are stored in DynamoDB as queued events. Admins can review, edit, approve, or reject them before publication.
- **Website**
Student calendar displays approved events; Admin Console manages event and organization review workflows.

IMPLEMENTED

- ✓ Automated multi-organization web scraping
- ✓ AI-powered event extraction with Amazon Bedrock
- ✓ Event de-duplication engine
- ✓ Admin approval workflows for events & organizations
- ✓ Student calendar connected to approved live event data

PROPOSED / NEXT

- Automated daily scraping at 6 AM
- Improve extraction accuracy across diverse websites
- Production deployment, monitoring & scalability

Demo calendar is seeded with realistic sample data modeled on 10 real partner orgs, not live-scraped -- see next slide.



IMPACT FOR DELAWARE

Less staff time. More insights into real youth opportunities.



Families & youth

One place to search and filter,
instead of checking dozens of sites



Partner organizations

More visibility for their programs,
no manual double-posting



State agencies

A single, standing view of youth
events statewide

WHAT A DELAWARE AGENCY WOULD NEED TO PILOT THIS

- 1 DDOL sponsorship + a small API budget
- 2 Partner sign-off on scraping their public event pages
- 3 A reviewer to approve Pending submissions
- 4 Hosting on state web infrastructure or a simple cloud host



RESPONSIBLE AI & DATA

Nothing goes live without a human confirmation

Data used

Only pull data from publicly available, open access website that are provided our partner organizations

Privacy & security

No scraping of private or gated pages. Submitted contact details are visible only to signed-in admins

AI limitations

The model can misread ambiguous dates or costs. It's instructed to output null rather than guess

Human oversight

Every AI-extracted or crowd-submitted event lands as Pending. Details are listed, so a an admin can make the final decision

Beyond the API call

We built the scraping pipeline to create one event id and hash maps and checks if there is the same id or not



PLATFORM & LEARNING

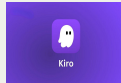
Platform / services used: *AWS Bedrock, Lambda functions, Step functions, VS Code, Kiro, ChatGPT, Claude*

What actually ran on it: *AWS host our web scraper and express Javascript backend, DynamoDB host all data for website*

Why this platform: *We found found the json like model of DynamoDB for handling data, and found it was easier to stay within the ecosystem*

What worked / didn't: *Very friendly eco system, that allowed kiro to put our project architecture on their systems*

What we learned: *Dedicate more time to preplaning our architecture, how to use multiple services in one project*





DEMO / RESULTS

Input → processing → output, in one pass

INPUT

"...join us for our Back-to-School Backpack Give-Away, Sun 8/23, 10am-1pm at H. Fletcher Brown Club..."

raw text scraped from a partner page

PROCESSING

Web Scraper extracts against a strict schema: subject, startDate, endDate, location, cost, confidence

one call per source

OUTPUT

```
export type Event = {  
  org_name: string,  
  link: string,  
  eventId: string,  
  startDate: string,  
  endDate: string,  
  location: string,  
  age_range: string,  
  cost: string,  
  description: string,  
  subject: string  
}
```

structured JSON, ready for the calendar

STRAIGHT FROM THE WEBSITES EVENT QUEUE

Hatch Chile Fest

Wegmans

Date/time not provided

No location provided

Missing start/end date or time

Missing location

Edit



700+

sample events over

100 real partner orgs



FROM PROTOTYPE TO PRODUCTION

Works today

- Scraper extraction
- Deduplication logic
- Publicly Downloadable Calendar
- Full Admin Control

Current limits

- No live automation
- No easy admin manual refresh
- No mailing list scraping
- No alternate scraping routine

Needed for production

- Automated Web Scraping
- Smart event confirmation
- Scheduled specific site rescrapes

Next steps

- Deploy to a state-hosted domain
- Have to add state login for frontend and backend
- Live Subscribable Calendar

Takeaway

PROBLEM

130+ organizations,
130+ calendars



SOLUTION

One trusted calendar



RESULT

Working scraper + extraction



IMPACT

Easy Access for all, Helps State
distribute resources efficiently

No youth misses a real opportunity because no one saw the flyer.